

Corso di Laurea Magistrale in Digital Transformation Management

Fancy Title

Tesi di laurea in:
BUSINESS INTELLIGENCE

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Abstract

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Chapter 1

Introduction

1.1 Context

During one of the audited weeks of my internship, the operational dashboard flagged a clear productivity gap at the station. The AI summary built on top of it explained the gap with full confidence, but the explanation produced was factually incorrect. It pointed to a structural metric that had little to do with the real cause of the difference, proposed corrective actions and root analyses that no manager could realistically carry out. I noticed the error, that morning, because i was familiar with the station and the metric, but a manager that reads the same paragraph on a busy day or even if less experienced could have easily take that for true and acted on it.

That episode marked my thesis begin and frames the general problem it addresses: what happen when a Large Language Model (LLM) is placed on top of a Business Intelligence (BI) dashboard and asked to turn numbers into decisions? That's a key question considering the setting of a Delivery Station (DS). It's the final node of the logistick network, the place where package are sorted in the early morning and loaded onto routes to be delivered to final customers. The station where i worked, DFV2 in Udine, runs on a standard daily rhythm: plan the day, run it, measure what happened, adjust for tomorrow. Manager controls the rhythm trough a large set of indicators that read every day to determine if any correction is needed.

Normally the window to act is narrow so is important that a Manager doesn't get the picture late or even wrong otherwise the chances to fix it would be lost.

Among all the indicators that we used at the station, one productivity ratio act as the headline figure, it's a number that represent the capability of the station to turn paid work into delivered packages. Most of the days this ratio is close to the planned one, but not exactly on it and that gap is what a manager has to explore, understand and explain. A process that is anything but trivial. This difference is the combined effect of many separated factors, some of them are internal to the station while the rest of them is completely extrarnal, and working out which factor and its weight is a task that, before this project, was done completely by hand.

When a station miss its target, the designed manager has to produce a written justification , internally refferred as the "bridge", based on a weekly Excel report that states how the previous one had closed. A manual and slow activity that neither said anything about how the current week was going while there was still time to act on it.

This is the exact scenario where a LLM could be the right fit. A that model is capable of reading the numbers and turn them into praragraphs within seconds, replacing hours of manual assembly. The downside is that the same flexibility that makes it so powerfull and usefull is also the source of risks. A language model hallucinates on numbers, quietly recomputing a value or inventing one that looks plausible, and it explains causes with the same fluency whether the cause is real or not. For a tool that is used to feed decisions on the floor those errors are not acceptable and they must be reduced as much as possible, while keeping in mind that hallucinations cannot be avoided at 100%.

A manager must be able to safely rely on the output of the model while, at the same time have a clear view on the related data. That's why I have built a dynamic dashboard that provides on-demand, intra-week visibility for all the factors influencig the gap. It automatically retrieves the required data to avoid manual checks, while establishing the strict architectural constraints necessary to keep the model anchored to the verified numbers.

1.2 Criticalities

The main difficulty of this project is the grounding problem, intended in a stricter sense than the one commonly used in the literature. It can be decomposed into two main requirements that need to hold together. A numerical: every quantitative token in the generated text must be traceable, character by character, to the pre-computed value. The second one is causal: every explanation and every recommendation must come from a well defined set of causes that the domain admits, not from everything that seems reasonable.

Consider the numerical side first. If the value that the dashboard shows is 49.0, the sentence produced by the LLM must say 49.0, not 49 or "around 50". This is more demanding than the usual faithfulness a retrieval system offers, where a claim counts as grounded if it is broadly supported by some documentation. For Operational BI a broad support is not enough. The number that is shown to the final user and the one that is retrieved by the code should be the same because otherwise any decision that is taken solely based on the Artificial Intelligence (AI) generated text when a figure is close but not exact can potentially, in this context, be wrong.

The causal side is even harder. Reporting a number without changing its value is a matter of fidelity to the base data. Explaining why we have that precise value, instead, is connected to the domain knowledge. The set of plausible-sounding causes for any gap is enormous, while the possible set of causes admitted by the domain is small and specific. A narrative defined as "useful" has to stay inside this small set. It should behave like an expert analyst that among everything that could explain the gap, choose only the feasible reasons and consequently actions that a Manager should take to improve the situation. Domain constraint without numerical fidelity buys very little, a right explanation based on a wrong input is meaningless.

1.3 Framework

The primary research question and the supporting sub-questions that guide the entire thesis aim to address how an AI-BI data binding framework can be designed to guarantee such fidelity.

Structure of the Thesis The remainder of this document is organized as follows:

- **Chapter 2** reviews the state of the art and related works, exploring current paradigms for data integration, evaluating hallucination metrics, and identifying the specific research gap in operational BI.
- **Chapter 3** introduces the case study, detailing the operational setting, the core metric, the pre-AI baseline, and the single-case methodology adopted during the internship.
- **Chapter 4** presents the proposed solution, detailing the conceptual innovation of the three-contract framework (tile-as-source, curated knowledge tree, and sanitizer) and the system architecture.
- **Chapter 5** focuses on the results and evaluation, analyzing the catalog of grounding failures and providing both quantitative controlled assessments and real-world qualitative evidence.
- **Chapter 6** discusses the findings, highlighting the thesis contributions and generalizability, acknowledging limitations, and outlining future research directions.

Chapter 2

State of the Art and Related Works

2.1 Data Integration

Main approaches for connecting LLMs to data, including Retrieval-Augmented Generation (RAG), prompt-based methods, tool use, and domain knowledge injection via semantic layers.

2.2 Evaluation

Existing metrics used to measure and evaluate AI hallucinations.

2.3 Literature

Current research and advancements in GenBI and LLM OLAP.

2.4 Research Gap

Identification of the specific gap in the current literature that this thesis aims to bridge by proposing a unified framework.

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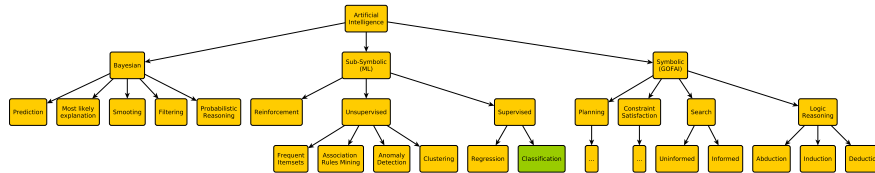


Figure 2.1: Some random image

Chapter 3

Case Study: Context and Problem

3.1 Setting

The last-mile delivery station environment, the continuous improvement cycle (plan, execute, measure, adjust), and the time pressure that makes delayed data visibility a missed intervention window.

3.2 Core Metric

Definition of the reference indicator as a productivity ratio (output units divided by input resource), the gap between planned and actual values, and its additive decomposition into a fixed set of mutually-exclusive categories with a two-level hierarchy, heterogeneous aggregation weights, and a residual term that guarantees completeness.

3.3 Pre-AI Baseline

The manual workflow for monitoring (weekly cadence, tile-by-tile extraction) and for producing the bridge document (subjective cause selection, unconstrained ranking, experience-dependent actions).

3.4 Three Weaknesses

Limited reactivity, subjective root-cause selection, and analyst dependence, mapped explicitly to the three contracts proposed in Chapter 4.

3.5 Methodology

Single-case longitudinal design where the researcher is embedded in the operational team for six months. Development followed an iterative pattern: observe a failure in the AI output, isolate it, fix it, validate against real data, check for regressions. Failure instances were collected and classified into families for the analysis in Chapter 5. A synthetic dataset preserving the structural properties of the real data built for the controlled evaluation.

Chapter 4

Proposed Solution: Three-Contract Framework

4.1 Core Concept

Introduction of the three contracts (tile-as-source, curated knowledge tree, and post-generation sanitizer) as a unified solution to the grounding problem, each acting at a different point of the generation chain.

4.2 Prompt Engineering

The design of the system prompt as an explicit encoding of fidelity rules, with one directive per observed failure mode rather than generic accuracy instructions.

4.3 Architecture

The overall system architecture of the AI-BI pipeline, from data aggregation to final narrative output.

```
1 public class HelloWorld {  
2     public static void main(String[] args) {  
3         // Prints "Hello, World" to the terminal window.  
4         System.out.println("Hello, World");  
5     }  
6 }
```

4.4 Application

How the framework was derived from the iterative development of the case study, showing the concrete implementation of each contract.

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4.5 Fancy formulas here

Chapter 5

Results and Evaluation

5.1 Failure Analysis

A catalog of the main grounding failure modes observed and an explanation of how the proposed framework mitigates them.

5.2 Quantitative Assessment

A controlled evaluation using mock data to compare different system configurations.

5.3 Qualitative Assessment

Two to three real-world qualitative episodes drawn from the case study to provide practical, operational evidence of success.

Chapter 6

Discussion and Future Directions

6.1 Synthesis

How the observed results successfully answer the initial research question.

6.2 Contributions

The specific value added to the existing literature regarding LLMs and BI.

6.3 Generalizability

An assessment of how well the three-contract framework can be adapted and applied to other contexts or industries.

6.4 Limitations

A transparent look at the limits of the current thesis work.

6.5 Future Developments

Potential next steps, such as extending the framework to support interactive Q&A functionalities based on the same underlying schema.

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